



Blacktown Boys' High School

2018

HSC Trial Examination

Mathematics Extension 2

**General
Instructions**

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- NESA approved calculators may be used
- A reference sheet is provided for this paper
- All diagrams are not drawn to scale
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

**Total marks:
100****Section I – 10 marks** (pages 3 – 6)

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 90 marks (pages 7 – 12)

- Attempt Questions 11 – 16
- Allow about 2 hours and 45 minutes for this section

Assessor: X. Chirgwin

Student Name: _____

Teacher Name: _____

Students are advised that this is a trial examination only and cannot in any way guarantee the content or format of the 2018 Higher School Certificate Examination.

Section I**10 marks****Attempt Questions 1–10**

Use the multiple choice answer sheet provided on page 14 for Questions 1–10.

1 Which conic has eccentricity $\frac{\sqrt{21}}{5}$?

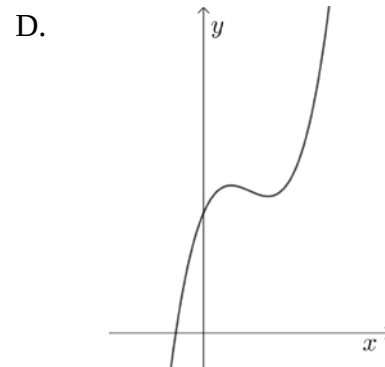
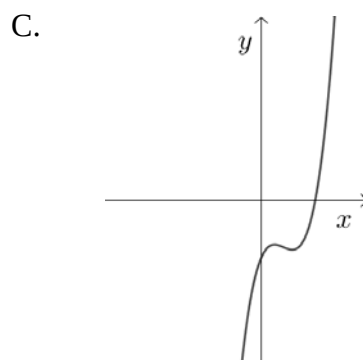
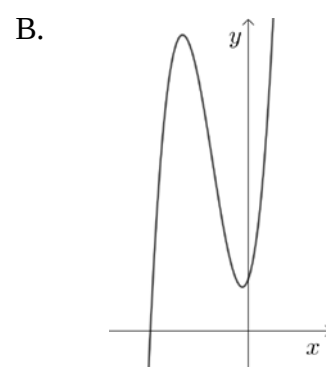
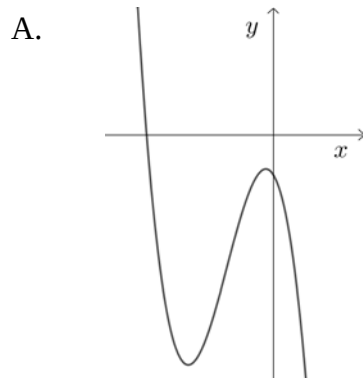
A. $\frac{x^2}{5^2} + \frac{y^2}{2^2} = 1$

B. $\frac{x^2}{5} + \frac{y^2}{2} = 1$

C. $\frac{x^2}{5^2} - \frac{y^2}{2^2} = 1$

D. $\frac{x^2}{5} - \frac{y^2}{2} = 1$

2 The polynomial $P(x) = x^3 - 5x^2 + 7x + 13$ has a zero at $x = 3 + 2i$. Which of the graphs below could be the graph of $P(x)$?



3 What value of z satisfies $z^2 = 32 + 24i$?

A. $2 + 6i$

B. $2 - 6i$

C. $6 + 2i$

D. $6 - 2i$

4 If w is a complex root of $z^5 - 1 = 0$, then $(1 + w - w^2 + w^3 + w^4)^{2018}$ is equal to

A. $2^{2018}w^3$

B. $2^{2018}w^2$

C. $2^{2018}w$

D. 2^{2018}

5 The primitive function of $\cos^3 x$ is

A. $\sin x - \frac{1}{3}\sin^3 x + C$

B. $\frac{1}{3}\sin^3 x - \sin x + C$

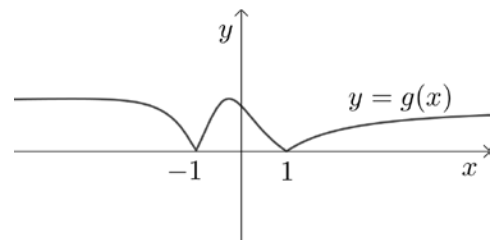
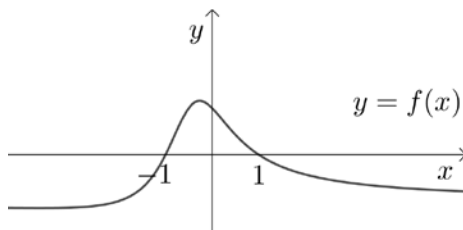
C. $\frac{1}{4}\sin^4 x + C$

D. $\frac{1}{4}\cos^4 x + C$

- 6 Consider the region bounded by the y -axis, the line $y = 3$ and the curve $y = e^x$. If the region is rotated about the line $y = 3$, which expression gives the volume of the solid of revolution?

- A. $V = \pi \int_1^3 (3 - e^x)^2 dx$
- B. $V = \pi \int_0^{\ln 3} (3 - e^x)^2 dx$
- C. $V = \pi \int_1^3 (3 - y) dy$
- D. $V = \pi \int_0^{\ln 3} e^{2x} dx$

- 7 Which statement is true for these graphs?



- A. $g(x) = \sqrt{f(x)}$
- B. $|f(x)| = g(x)$
- C. $g(x) = [f(x)]^{-1}$
- D. $f(\sqrt{x}) = g(x)$

- 8** Suppose that $f(x)$ is a non-zero continuous odd function. Which of the statement below is true?

A. $\int_{-1}^1 (f(x) \cos x) dx = 2 \int_0^1 (f(x) \cos x) dx$

B. $\int_{-1}^1 (f(x^2) \sin x) dx = 2 \int_0^1 (f(x^2) \sin x) dx$

C. $\int_{-1}^1 (f(x) \tan x) dx = 2 \int_0^1 (f(x) \tan x) dx$

D. $\int_{-1}^1 (f(x^2) e^x) dx = 2 \int_0^1 (f(x^2) e^x) dx$

- 9** A committee of three to be chosen at random from 6 women and n men ($n \geq 1$). Find the number of possible committees containing exactly one women.

A. nC_2

B. 20

C. $6n$

D. $3n(n-1)$

- 10** The normal at the point $P\left(cp, \frac{c}{p}\right)$ on the rectangular hyperbola $xy = c^2$ has equation $p^3x - py = cp^4 - c$. This normal cuts the hyperbola at a second point $Q\left(cq, \frac{c}{q}\right)$.

What is the relationship between p and q ?

A. $p^3q = -1$

B. $pq^4 = -1$

C. $p^2q = -1$

D. $pq = -1$

End of Section I

Section II**90 Marks****Attempt Questions 11–16**

Answer each question on a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11–16, your responses should include relevant mathematical reasoning and/or calculations.

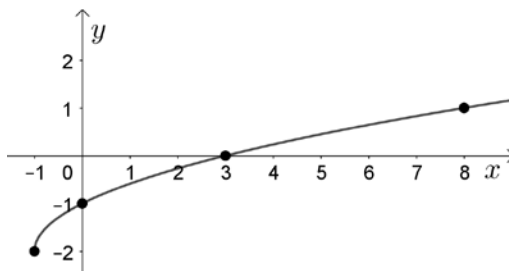
Question 11 (15 marks) Use a SEPARATE writing booklet.

- a) Let $z = \sqrt{3} - i$ and $w = 4\sqrt{3} + 2i$
- i) Find $\bar{w} + z^{-1}$ in the form $a + bi$, where a and b are real. 2
 - ii) Express z in modulus-argument form. 2
 - iii) Write z^{18} in its simplest form. 2
- b) i) Find the numbers A, B and C such that 2
- $$\frac{2x^2 + 9x + 1}{(x - 2)(x^2 + 5)} = \frac{A}{x - 2} + \frac{Bx + C}{x^2 + 5}$$
- ii) Hence, or otherwise, find 2
- $$\int \frac{2x^2 + 9x + 1}{(x - 2)(x^2 + 5)} dx$$
- c) Sketch $\frac{x^2}{16} + \frac{y^2}{11} = 1$ indicating the coordinates of its foci and equations 2
of the directrices.
- d) Find the value of $\frac{dy}{dx}$ at the point $(-1, 5)$ on the curve $x^3y^2 = 10 + y$. 3

End of Questions 11

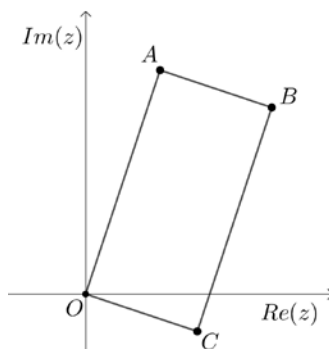
Question 12 (15 marks) Use a SEPARATE writing booklet.

- a) The diagram below shows the graph of the function $f(x) = \sqrt{x+1} - 2$.



On separate diagrams sketch the following graphs, showing clearly any intercepts on the coordinate axes, key points, and the equations of any asymptotes:

- | | | |
|------|----------------------|---|
| i) | $y = f(x) $ | 1 |
| ii) | $y = [f(x)]^2$ | 2 |
| iii) | $y^2 = f(x)$ | 2 |
| iv) | $y = f(x^2)$ | 2 |
| v) | $y = \frac{1}{f(x)}$ | 2 |
| vi) | $y = x \cdot f(x)$ | 2 |
| vii) | $y = \log_e[f(x)]$ | 2 |
- b) $OABC$ is a rectangle on the Argand diagram shown below, where O is the origin. If A represents the complex number $2 + 6i$, find the complex numbers represented by B and C , given that the side OC is half the length of OA . 2



End of Questions 12

Question 13 (15 marks) Use a SEPARATE writing booklet.

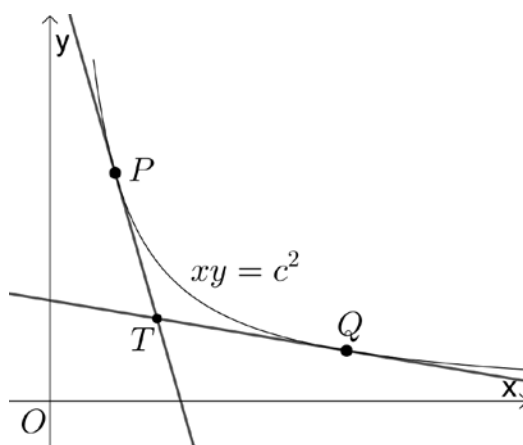
- a) If α, β and γ are the roots of the polynomial equation $2x^3 - 7x + 3 = 0$, form the polynomial equation with roots:
- i) $\alpha + 1, \beta + 1$, and $\gamma + 1$ 2
 - ii) α^{-2}, β^{-2} , and γ^{-2} 2
- b) Consider the hyperbolas $H_1: \frac{x^2}{9} - \frac{y^2}{16} = 1$ and $H_2: \frac{y^2}{16} - \frac{x^2}{9} = 1$.
- i) Show that the foci of both hyperbolas lies on the same circle. 2
 - ii) Hence, or otherwise, write down the equation of this circle. 1
- c) Evaluate $\int_0^a x \sin(a - x) dx$ 3
- d) i) Find, in modulus-argument form, the three cube roots of -64 . 2
- ii) Write the two unreal cube roots of -64 in the form $a + bi$, where a and b are real. 1
- iii) If w_1 and w_2 are the unreal cube roots of -64 , show that 2
- $$w_1^{6n} + w_2^{6n} = 2^{12n+1}$$
- for all integers n .

End of Questions 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- a) Find, by the method of cylindrical shells, the volume of the solid generated when the region bounded by the curve $y = x^2 + 3$, the lines $x = 2$ and the x and y axes is rotated about the line $x = 4$. 3

- b) The distinct points $P\left(cp, \frac{c}{p}\right)$ and $Q\left(cq, \frac{c}{q}\right)$ are on the same branch of the Hyperbola with the equation $xy = c^2$. The tangents at P and Q meet at the point T .



- i) Show that the equation of the tangent at P is $+p^2y = 2cp$. 2
- ii) Show that T has coordinates $\left(\frac{2cpq}{p+q}, \frac{2c}{p+q}\right)$. 2
- iii) Let P and Q move so that the tangent at P intersects the x -axis at $(cq, 0)$. Show that the locus of T is a hyperbola and state its eccentricity. 4
- c) A sequence of numbers T_n , $n = 1, 2, 3, \dots$ is defined by $T_1 = 2, T_2 = 0$ and $T_n = 2T_{n-1} - 2T_{n-2}$ for $n = 3, 4, 5, \dots$. 4
 Use mathematical induction to show that

$$T_n = (\sqrt{2})^{n+2} \cos\left(\frac{n\pi}{4}\right), n = 1, 2, 3, \dots$$

End of Questions 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- a) A chocolate has a circular base of radius 2 cm. If every section perpendicular to this base is an equilateral triangle, find the volume of chocolate needed to make a box of 30 such chocolates. 3

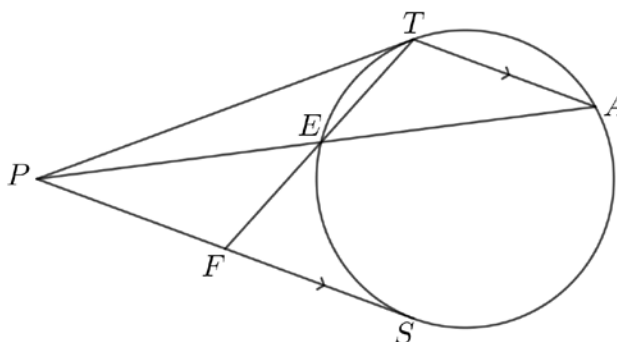
b) i) Show that $\frac{(1-t)^n}{t} = t^{n-1} \left(\frac{1}{t} - 1 \right)^n$ 1

ii) Let $I_n = \int_1^x \frac{(1-t)^n}{t} dt$, $n = 1, 2, 3, \dots$ 3

Show that $I_n = \frac{(1-x)^n}{n} + I_{n-1}$

- c) If a root of the cubic equation $ax^3 + bx^2 + cx + d = 0$ is equal to the reciprocal of another root, show that 3
- $$a^2 - d^2 = ac - bd$$

- d) The diagram below shows two tangents PT and PS drawn to a circle from a point P outside the circle. Through T , a chord TA is drawn parallel to the tangent PS . The secant PA meets the circle at E , and TE produced meets PS at F .



- i) Prove that $\triangle EFP$ is similar to $\triangle PFT$. 2
- ii) Hence show that $PF^2 = TF \times EF$. 1
- iii) Hence, or otherwise, prove that F is the midpoint of PS . 2

End of Questions 15

Question 16 (15 marks) Use a SEPARATE writing booklet.

- a) Four letters are chosen from the letters of the word TELEGRAPH. 2

These four letters are then placed alongside one another to form a four letter arrangement. Find the number of distinct four letter arrangements which are possible, considering all choices.

- b) If $m > 0, n > 0, p > 0, q > 0$, show that

i) $m + n \geq 2\sqrt{mn}$. 1

ii) $(m + n)(n + p)(m + p) \geq 8mnp$. 2

iii) $\frac{m}{n} + \frac{n}{p} + \frac{p}{q} + \frac{q}{m} \geq 4$. 2

- c) Given that $f(x) = x^{x^2}$, show that $f'(x) = (2 \log_e x + 1)x^{x^2+1}$. 2

- d) It is given that $A > 0, B > 0$ and n is a positive integer.

i) Simplify $\frac{A^{n+1} - A^n B + B^{n+1} - B^n A}{A - B}$. 1

ii) Deduce that $A^{n+1} + B^{n+1} \geq A^n B + B^n A$. 2

iii) Show by mathematical induction, that $\left(\frac{A+B}{2}\right)^n \leq \frac{A^n + B^n}{2}$. 3

End of Paper

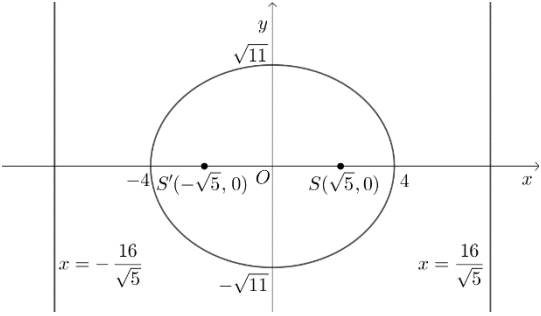
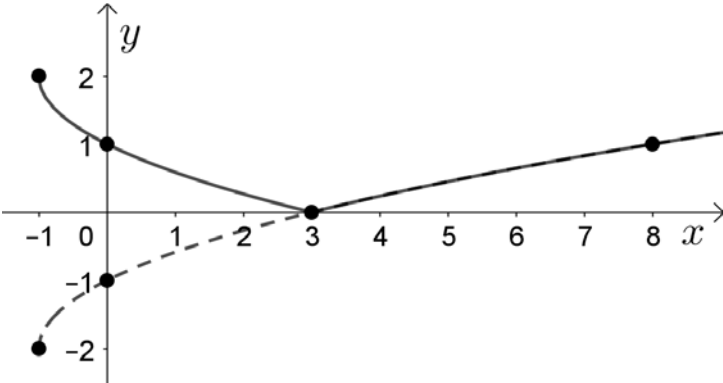
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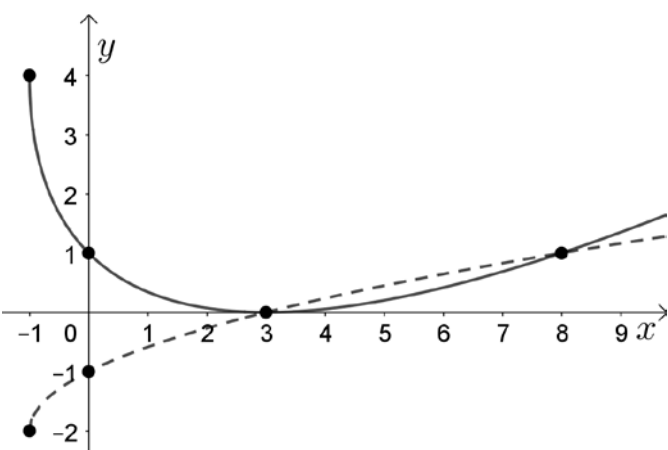
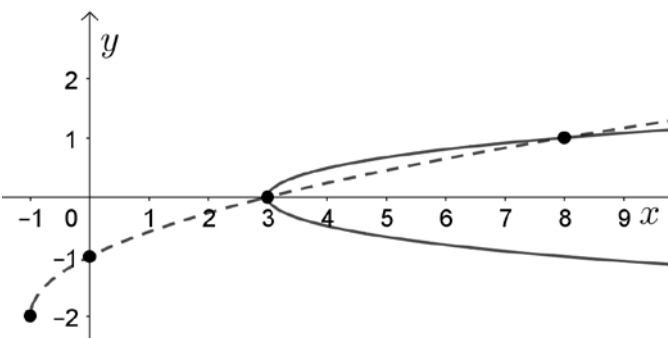
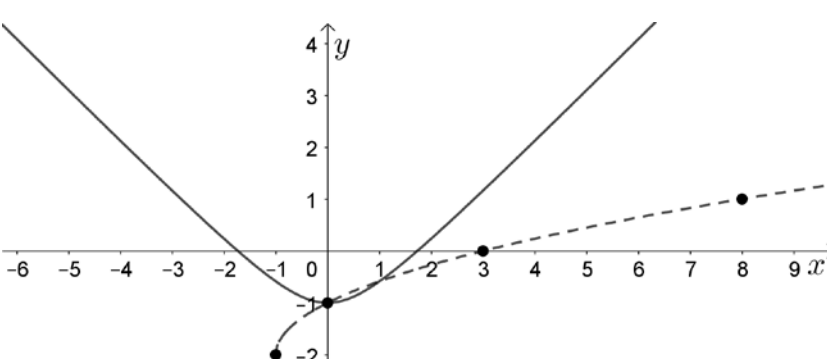
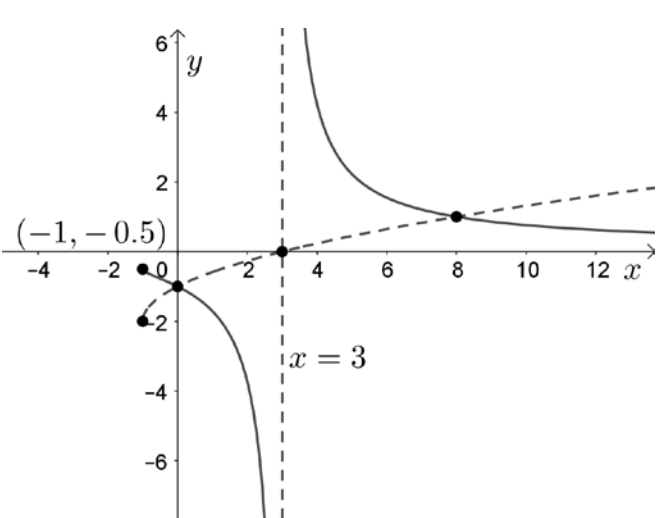
Section 1

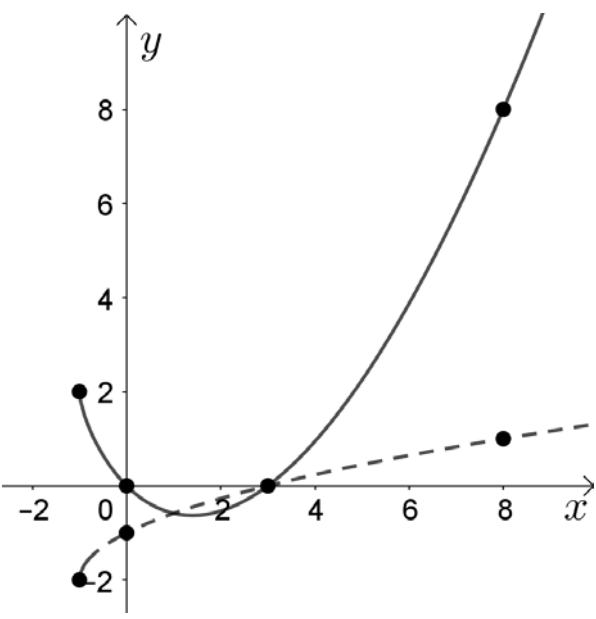
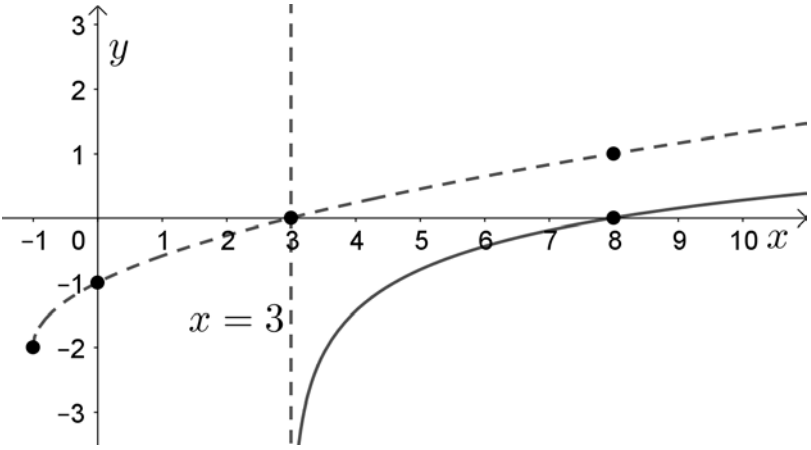
1	A Ellipse since $0 < e < 1$ $b^2 = a^2(1 - e^2)$ $\frac{b^2}{a^2} = 1 - \left(\frac{\sqrt{21}}{5}\right)^2$ $\frac{b^2}{a^2} = \frac{4}{25}$ $\therefore \frac{x^2}{5^2} + \frac{y^2}{2^2} = 1$	1 Mark
2	D The graph has a positive y-intercept so only possibility is B or D. Solving for stationary points, $P'(x) = 0$ $3x^2 - 10x + 7 = 0$ $(x - 1)(3x - 7) = 0$ $x = 1, x = \frac{7}{3}$ Hence, stationary points occurs in the first quadrant.	1 Mark
3	C $(6 + 2i)^2$ $= 36 + 24i - 4$ $= 32 + 24i$	1 Mark
4	C $(1 + w - w^2 + w^3 + w^4)^{2018}$ $= (1 + w + w^2 + w^3 + w^4 - 2w^2)^{2018}$ $= (-2w^2)^{2018} \quad (1 + w + w^2 + w^3 + w^4 = 0)$ $= 2^{2018} w^{4036}$ $= 2^{2018} w^{4035} w$ $= 2^{2018} w$	1 Mark
5	A $\int \cos^3 x \, dx$ $= \int \cos x \cos^2 x \, dx$ $= \int \cos x (1 - \sin^2 x) \, dx$ $= \int (\cos x - \cos x \sin^2 x) \, dx$ $= \sin x - \frac{1}{3} \sin^3 x + C$	1 Mark
6	B $\Delta V = \pi(3 - y)^2 \Delta x$ $V = \pi \int_0^{\ln 3} (3 - y)^2 \, dx$ $V = \pi \int_0^{\ln 3} (3 - e^x)^2 \, dx$	1 Mark
7	B $ f(x) = g(x)$	1 Mark

8	C Odd $\times$ Odd = Even $\therefore \int_{-1}^1 (f(x) \tan x) dx = 2 \int_0^1 (f(x) \tan x) dx$	1 Mark
9	D 1 woman from 6 women and 2 men from n men ${}^6C_1 \times {}^nC_2 = 6 \times \frac{n!}{(n-2)! 2!}$ ${}^6C_1 \times {}^nC_2 = 3n(n-1)$	1 Mark
10	A Q lies on $p^3x - py = cp^4 - c$ $Q\left(cq, \frac{c}{q}\right)$ $p^3 \times cq - p \times \frac{c}{q} = cp^4 - c$ $cp^3q - \frac{cp}{q} = cp^4 - c$ $p^3q^2 - p = p^4q - q$ $p^3q^2 - p^4q = p - q$ $p^3q(q - p) = -(q - p)$ $p^3q = -1$	1 Mark

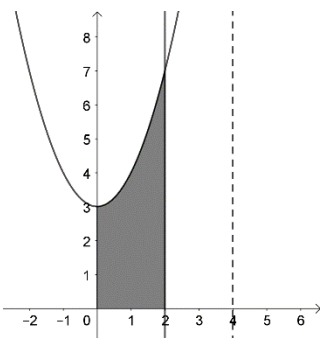
Section 2		
Q11 a) i)	$\bar{w} + z^{-1}$ $= 4\sqrt{3} - 2i + \frac{1}{\sqrt{3} - i} \times \frac{\sqrt{3} + i}{\sqrt{3} + i}$ $= 4\sqrt{3} - 2i + \frac{\sqrt{3} + i}{3 + 1}$ $= \frac{17\sqrt{3}}{4} - \frac{7}{4}i$	2 Marks Correct solution 1 Mark Obtains the correct expressions for $\bar{w}$ or z^{-1}
Q11 a) ii)	$z = \sqrt{3} - i$ $ z = \sqrt{3 + 1} = 2$ $\arg z = \tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = -\frac{\pi}{6}$ $z = 2\left(\cos\left(-\frac{\pi}{6}\right) + i\sin\left(-\frac{\pi}{6}\right)\right)$ $z = 2\left(\cos\frac{\pi}{6} - i\sin\frac{\pi}{6}\right)$	2 Marks Correct solution 1 Mark Finds modulus or argument
Q11 a) iii)	By de Moivre's theorem and part ii) z^{18} $= 2^{18}\left(\cos\left(18 \times \frac{\pi}{6}\right) - i\sin\left(18 \times \frac{\pi}{6}\right)\right)$ $= 2^{18}(\cos 3\pi - i\sin 3\pi)$ $= -2^{18}$	2 Marks Correct solution 1 Mark Applies de Moivre's theorem
Q11 b) i)	$A(x^2 + 5) + (Bx + C)(x - 2) = 2x^2 + 9x + 1$ Let $x = 2$ $A \times (2^2 + 5) + 0 = 2 \times 2^2 + 9 \times 2 + 1$ $9A = 27$ $A = 3$ Let $x = 0$ $5A - 2C = 1$ $5 \times 3 - 2C = 1$ $-2C = -14$ $C = 7$ Let $x = 1$ $6A + (B + C)(1 - 2) = 2 + 9 + 1$ $6 \times 3 + (B + 7) \times -1 = 12$ $-B - 7 = -6$ $-B = 1$ $B = -1$	2 Marks Correct solution 1 Mark Find the correct value of A or B or C with working
Q11 b) ii)	$\int \frac{2x^2 + 9x + 1}{(x - 2)(x^2 + 5)} dx$ $= \int \left(\frac{3}{x - 2} + \frac{-x + 7}{x^2 + 5} \right) dx$ $= \int \frac{3}{x - 2} dx - \frac{1}{2} \int \frac{2x}{x^2 + 5} dx + \int \frac{7}{x^2 + 5} dx$ $= 3 \ln x - 2 - \frac{1}{2} \ln x^2 + 5 + \frac{7}{\sqrt{5}} \tan^{-1} \frac{x}{\sqrt{5}} + C$ $= \ln \left \frac{(x - 2)^3}{\sqrt{x^2 + 5}} \right + \frac{7}{\sqrt{5}} \tan^{-1} \frac{x}{\sqrt{5}} + C$	2 Marks Correct solution 1 Mark Attempts to integrate using partial fractions and makes significant progress

Q11 c)	$\frac{x^2}{16} + \frac{y^2}{11} = 1$ $b^2 = a^2(1 - e^2)$ $11 = 16(1 - e^2)$ $\frac{11}{16} = 1 - e^2$ $e^2 = 1 - \frac{11}{16}$ $e = \frac{\sqrt{5}}{4}$ Foci $(\pm ae, 0)$ $= \left(\pm 4 \times \frac{\sqrt{5}}{4}, 0 \right)$ $= (\pm\sqrt{5}, 0)$ Equations of directrices: $x = \pm \frac{a}{e}$ $x = \pm \frac{4}{\frac{\sqrt{5}}{4}}$ $x = \pm \frac{16}{\sqrt{5}}$ 	2 Marks Correct solution and graph 1 Mark Finding the correct the foci or equations of directrices
Q11 d)	$x^3y^2 = 10 + y$ $3x^2y^2 + x^3 \times 2y \frac{dy}{dx} = \frac{dy}{dx}$ $3x^2y^2 = \frac{dy}{dx} - 2yx^3 \frac{dy}{dx}$ $\frac{dy}{dx}(1 - 2yx^3) = 3x^2y^2$ $\frac{dy}{dx} = \frac{3x^2y^2}{1 - 2yx^3}$ At $(-1, 5)$ $\frac{dy}{dx} = \frac{3 \times (-1)^2 \times 5^2}{1 - 2 \times 5 \times (-1)^3}$ $\frac{dy}{dx} = \frac{75}{11}$	3 Marks Correct solution 2 Mark Correct differentiation 1 Mark Attempts to differentiate implicitly using the chain rule
Q12 a) i)	$y = f(x) $ 	1 Mark Correct graph

Q12 a) ii)	$y = [f(x)]^2$ 	2 Marks Correct graph 1 Mark Correct key points indicated
Q12 a) iii)	$y^2 = f(x)$ 	2 Marks Correct graph 1 Mark Recognises correct domain
Q12 a) iv)	$y = f(x^2)$ 	2 Marks Correct graph 1 Mark Recognises reflection along the y-axis
Q12 a) v)	$y = \frac{1}{f(x)}$ 	2 Marks Correct graph 1 Mark Recognises and labels the asymptote

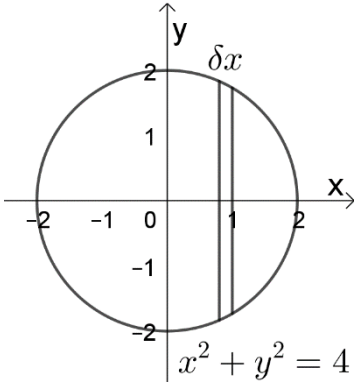
Q12 a) vi)	$y = x \cdot f(x)$ 	2 Marks Correct graph 1 Mark Correct key points indicated
Q12 a) vii)	$y = \log_e[f(x)]$ 	2 Marks Correct graph 1 Mark Recognises correct domain
Q12 b)	$\overrightarrow{OC} = \frac{1}{2} \times \overrightarrow{OA} \times (-i)$ $\overrightarrow{OC} = \frac{1}{2} \times (2 + 6i) \times (-i)$ $\overrightarrow{OC} = 3 - i$ $\overrightarrow{OB} = \overrightarrow{OA} + \overrightarrow{OC}$ $\overrightarrow{OB} = 2 + 6i + 3 - i$ $\overrightarrow{OB} = 5 + 5i$	2 Marks Correct solution 1 Mark Finds the complex number representing C
Q13 a) i)	$2x^3 - 7x + 3 = 0$ Let $x = \alpha + 1$ $\alpha = x - 1$ $2(x - 1)^3 - 7(x - 1) + 3 = 0$ $2(x^3 - 3x^2 + 3x - 1) - 7x + 7 + 3 = 0$ $2x^3 - 6x^2 - x + 8 = 0$	2 Marks Correct solution 1 Mark Recognises the relationships of the roots and makes the correct substitution

Q13 a) ii)	$2x^3 - 7x + 3 = 0$ Let $x = a^{-2} = \frac{1}{a^2}$ $a = \pm \frac{1}{\sqrt{x}}$ $2\left(\pm \frac{1}{\sqrt{x}}\right)^3 - 7\left(\pm \frac{1}{\sqrt{x}}\right) + 3 = 0$ $\pm \frac{2}{x\sqrt{x}} - \left(\pm \frac{7}{\sqrt{x}}\right) + 3 = 0$ $\pm \frac{1}{\sqrt{x}}\left(\frac{2}{x} - 7\right) = -3$ $\left[\pm \frac{1}{\sqrt{x}}\left(\frac{2}{x} - 7\right)\right]^2 = (-3)^2$ $\frac{1}{x}\left(\frac{4}{x^2} - \frac{28}{x} + 49\right) = 9$ $4 - 28x + 49x^2 = 9x^3$ $9x^3 - 49x^2 + 28x - 4 = 0$	2 Marks Correct solution 1 Mark Recognises the relationships of the roots and makes the correct substitution
Q13 b) i)	$H_1: \frac{x^2}{9} - \frac{y^2}{16} = 1$ $b^2 = a^2(e^2 - 1)$ $16 = 9(e^2 - 1)$ $e^2 = \frac{16}{9} + 1$ $e = \frac{5}{3}$ Foci of $H_1: (\pm ae, 0) = (\pm 5, 0)$ $H_2: \frac{y^2}{16} - \frac{x^2}{9} = 1$ $a^2 = b^2(e^2 - 1)$ $9 = 16(e^2 - 1)$ $e^2 = \frac{9}{16} + 1$ $e = \frac{5}{4}$ Foci of $H_2: (0, \pm be) = (0, \pm 5)$ The four foci are 5 units from the origin. $\therefore$ They all lie on the same circle with centre at origin and radius 5.	2 Marks Correct solution 1 Mark Provides the correct foci for both hyperbolas
Q13 b) ii)	$x^2 + y^2 = 25$	1 Mark Correct solution
Q13 c)	$I = \int_0^a x \sin(a - x) dx$ $u = x \quad v' = \sin(a - x)$ $u' = 1 \quad v = -\cos(a - x)$ $I = [x \cos(a - x)]_0^a - \int_0^a \cos(a - x) dx$ $I = [a \cos(a - a) - 0] - [\sin(a - x)]_0^a$ $I = a - \sin a$	3 Marks Correct solution 2 Marks Correct integration 1 Mark Attempts to integrate by parts

Q13 d) i)	$z^3 = -64$ $z^3 = 64(\cos \pi + i \sin \pi)$ $z^3 = 64(\cos(\pi + 2k\pi) + i \sin(\pi + 2k\pi)) \quad \text{where } k \text{ is an integer}$ $z = 64^{\frac{1}{3}} \left(\cos \left(\frac{\pi + 2k\pi}{3} \right) + i \sin \left(\frac{\pi + 2k\pi}{3} \right) \right)$ $z = 4 \left(\cos \left(\frac{\pi + 2k\pi}{3} \right) + i \sin \left(\frac{\pi + 2k\pi}{3} \right) \right)$ Let $k = -1, 0, 1$ The three cube roots of -64 are: $4 \left(\cos \left(-\frac{\pi}{3} \right) + i \sin \left(-\frac{\pi}{3} \right) \right),$ $4 \left(\cos \left(\frac{\pi}{3} \right) + i \sin \left(\frac{\pi}{3} \right) \right),$ and $4(\cos(\pi) + i \sin(\pi))$	2 Marks Correct solution 1 Mark Applies De Moivre's theorem
Q13 d) ii)	$4 \left(\cos \left(-\frac{\pi}{3} \right) + i \sin \left(-\frac{\pi}{3} \right) \right)$ $= 4 \left(\frac{1}{2} - \frac{\sqrt{3}}{2} i \right)$ $= 2 - 2\sqrt{3}i$ $4 \left(\cos \left(\frac{\pi}{3} \right) + i \sin \left(\frac{\pi}{3} \right) \right)$ $= 4 \left(\frac{1}{2} + \frac{\sqrt{3}}{2} i \right)$ $= 2 + 2\sqrt{3}i$	1 Mark Correct solution
Q13 d) iii)	Let $w_1 = 4 \left(\cos \left(-\frac{\pi}{3} \right) + i \sin \left(-\frac{\pi}{3} \right) \right)$, $w_2 = 4 \left(\cos \left(\frac{\pi}{3} \right) + i \sin \left(\frac{\pi}{3} \right) \right)$ $w_1^{6n} + w_2^{6n}$ $= w_1^{6n} + w_2^{6n}$ $= (w_1^3)^{2n} + (w_2^3)^{2n}$ $= (-64)^{2n} + (-64)^{2n}$ $= 64^{2n} + 64^{2n}$ $= (2^6)^{2n} + (2^6)^{2n}$ $= 2 \times 2^{12n}$ $= 2^{12n+1}$	2 Marks Correct solution 1 Mark Recognises the relation of roots
Q14 a)	$V = 2\pi \int_0^2 (4-x)y \, dx$ $V = 2\pi \int_0^2 (4-x)(x^2+3) \, dx$ $V = 2\pi \int_0^2 (4x^2+12-x^3-3x) \, dx$ $V = 2\pi \left[\frac{4x^3}{3} + 12x - \frac{x^4}{4} - \frac{3x^2}{2} \right]_0^2$ $V = 2\pi \left(\frac{4 \times 2^3}{3} + 12 \times 2 - \frac{2^4}{4} - \frac{3 \times 2^2}{2} \right)$ $V = \frac{148\pi}{3} \text{ units}^3$ 	3 Marks Correct solution 2 Marks Correct primitive function 1 Mark Applies cylindrical shell method

Q14 b) i)	$xy = c^2$ $y = \frac{c^2}{x}$ $\frac{dy}{dx} = -\frac{c^2}{x^2}$ $P\left(cp, \frac{c}{p}\right)$ $m_P = -\frac{c^2}{(cp)^2}$ $m_P = -\frac{1}{p^2}$ Equation of tangent at P $y - \frac{c}{p} = -\frac{1}{p^2}(x - cp)$ $p^2y - cp = -x + cp$ $\therefore x + p^2y = 2cp$	2 Marks Correct solution 1 Mark Obtains the correct gradient of tangent at point P
Q14 b) ii)	Similarly, equation of tangent at Q is $x + q^2y = 2cq$ $x + p^2y = 2cp \quad (1)$ $x + q^2y = 2cq \quad (2)$ (1) – (2) $(p^2 - q^2)y = 2c(p - q)$ $y = \frac{2c(p - q)}{(p^2 - q^2)}$ $y = \frac{2c(p - q)}{(p + q)(p - q)}$ $y = \frac{2c}{p + q}$ Substitute into (1) $x + p^2 \times \frac{2c}{p + q} = 2cp$ $x = 2cp - \frac{2cp^2}{p + q}$ $x = \frac{2cp(p + q)}{p + q} - \frac{2cp^2}{p + q}$ $x = \frac{2cp^2 + 2cpq - 2cp^2}{p + q}$ $x = \frac{2cpq}{p + q}$ $\therefore T\left(\frac{2cpq}{p + q}, \frac{2c}{p + q}\right)$	2 Marks Correct solution 1 Mark Obtains the correct x or y coordinates of T
Q14 b) iii)	Given $(cq, 0)$ lies on tangent at P: $x + p^2y = 2cp$ $cq + 0 = 2cp$ $q = 2p$ $T\left(\frac{2cpq}{p + q}, \frac{2c}{p + q}\right)$ $x = \frac{2cpq}{p + q}$	4 Marks Correct solution 3 Marks Finds the locus of T 2 Marks Obtains x and y in terms p.

	$x = \frac{2cp \times 2p}{p + 2p}$ $x = \frac{4cp^2}{3p}$ $x = \frac{4cp}{3}$ $y = \frac{2c}{p + q}$ $y = \frac{2c}{p + 2p}$ $y = \frac{2c}{3p}$ $xy = \frac{4cp}{3} \times \frac{2c}{3p}$ $xy = \frac{8c^2}{9}$ Since $\frac{8c^2}{9}$ is a constant, then $xy = \frac{8c^2}{9}$ is a rectangular hyperbola with eccentricity $\sqrt{2}$.	1 Mark Establishing the relationship of $q = 2p$
Q14 c)	$T_n = (\sqrt{2})^{n+2} \cos\left(\frac{n\pi}{4}\right)$ 1. Prove statement is true for $n = 1, n = 2$ $T_1 = (\sqrt{2})^{1+2} \cos\left(\frac{\pi}{4}\right)$ $T_1 = 2\sqrt{2} \times \frac{1}{\sqrt{2}}$ $T_1 = 2$ $T_2 = (\sqrt{2})^{2+2} \cos\left(\frac{2\pi}{4}\right)$ $T_2 = 4 \times 0$ $T_2 = 0$ $\therefore$ Statement is true for $n = 1$ and $n = 2$. 2. Assume statement is true for $n = k$ (k some positive integer) $T_k = (\sqrt{2})^{k+2} \cos\left(\frac{k\pi}{4}\right)$ 3. Prove statement is true for $n = k + 1$ i. e. $T_{k+1} = (\sqrt{2})^{k+3} \cos\left(\frac{(k+1)\pi}{4}\right)$ $T_{k+1} = 2T_k - 2T_{k-1}$ $T_{k+1} = 2 \times (\sqrt{2})^{k+2} \cos\left(\frac{k\pi}{4}\right) - 2 \times (\sqrt{2})^{(k-1)+2} \cos\left(\frac{(k-1)\pi}{4}\right)$ $T_{k+1} = (\sqrt{2})^{k+4} \cos\left(\frac{k\pi}{4}\right) - (\sqrt{2})^{k+3} \cos\left(\frac{(k-1)\pi}{4}\right)$ $T_{k+1} = (\sqrt{2})^{k+3} \left(\sqrt{2} \cos\left(\frac{k\pi}{4}\right) - \cos\left(\frac{k\pi}{4} - \frac{\pi}{4}\right) \right)$	4 Marks Correct solution 3 Marks Simplifies the T_{k+1} terms and attempts to apply the difference of angles for cosine 2 Marks Establish the statement for T_{k+1} 1 Mark Prove statement is true for $n = 1$ and $n = 2$

	$T_{k+1} = (\sqrt{2})^{k+3} \left(\sqrt{2} \cos\left(\frac{k\pi}{4}\right) - \left(\cos\left(\frac{k\pi}{4}\right) \cos\left(\frac{\pi}{4}\right) + \sin\left(\frac{k\pi}{4}\right) \sin\left(\frac{\pi}{4}\right) \right) \right)$ $T_{k+1} = (\sqrt{2})^{k+3} \left(\sqrt{2} \cos\left(\frac{k\pi}{4}\right) - \frac{1}{\sqrt{2}} \cos\left(\frac{k\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{k\pi}{4}\right) \right)$ $T_{k+1} = (\sqrt{2})^{k+3} \left(\frac{1}{\sqrt{2}} \cos\left(\frac{k\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{k\pi}{4}\right) \right)$ $T_{k+1} = (\sqrt{2})^{k+3} \left(\cos\left(\frac{\pi}{4}\right) \cos\left(\frac{k\pi}{4}\right) - \sin\left(\frac{\pi}{4}\right) \sin\left(\frac{k\pi}{4}\right) \right)$ $T_{k+1} = (\sqrt{2})^{k+3} \left(\cos\left(\frac{k\pi}{4} + \frac{\pi}{4}\right) \right)$ $T_{k+1} = (\sqrt{2})^{k+3} \left(\cos\frac{(k+1)\pi}{4} \right)$ $\therefore$ True by mathematical induction for all positive integers n.	
Q15 a)	$A = \frac{1}{2} \times 2y \times 2y \times \sin 60^\circ$ $A = \sqrt{3}y^2$ $\Delta V = \lim_{\Delta x \rightarrow 0} \sum_{x=-2}^2 \sqrt{3}y^2 \Delta x$ $V = \int_{-2}^2 \sqrt{3}y^2 dx$ $V = 2\sqrt{3} \int_0^2 (4 - x^2) dx$ $V = 2\sqrt{3} \times \left[4x - \frac{x^3}{3} \right]_0^2$ $V = 2\sqrt{3} \times \left(4 \times 2 - \frac{2^3}{3} \right)$ $V = \frac{32\sqrt{3}}{3} \text{ cm}^3$ $\therefore$ The total volume of 30 chocolates is $\frac{32\sqrt{3}}{3} \times 30 = 320\sqrt{3} \text{ cm}^3$ 	3 Marks Correct solution 2 Marks Makes significant progress in finding the volume 1 Mark Finds the cross-sectional area
Q15 b) i)	$LHS = \frac{(1-t)^n}{t}$ $LHS = \frac{\left[t \left(\frac{1}{t} - 1 \right) \right]^n}{t}$ $LHS = \frac{t^n \left(\frac{1}{t} - 1 \right)^n}{t}$ $LHS = \frac{t^n}{t^1} \left(\frac{1}{t} - 1 \right)^n$ $\therefore \frac{(1-t)^n}{t} = t^{n-1} \left(\frac{1}{t} - 1 \right)^n$	1 Mark Correct solution

Q15 b) ii)	$I_n = \int_1^x \frac{(1-t)^n}{t} dt$ $I_n = \int_1^x t^{n-1} \left(\frac{1}{t} - 1\right)^n dt$ $u = (t^{-1} - 1)^n \qquad v' = t^{n-1}$ $u' = n\left(\frac{1}{t} - 1\right)^{n-1} \times (-t^{-2}) \qquad v = \frac{t^n}{n}$ $I_n = \left[\frac{t^n}{n} \left(\frac{1}{t} - 1\right)^n \right]_1^x - \int_1^x n\left(\frac{1}{t} - 1\right)^{n-1} \times (-t^{-2}) \times \frac{t^n}{n} dt$ $I_n = \left[\frac{x^n}{n} \left(\frac{1}{x} - 1\right)^n - \frac{1^n}{n} \left(\frac{1}{1} - 1\right)^n \right] + \int_1^x t^{n-2} \left(\frac{1}{t} - 1\right)^{n-1} dt$ $I_n = \frac{x^n}{n} \left(\frac{1}{x} - 1\right)^n + I_{n-1}$ $I_n = \frac{x}{n} \times x^{n-1} \left(\frac{1}{x} - 1\right)^n + I_{n-1}$ $I_n = \frac{x}{n} \times \frac{(1-x)^n}{x} + I_{n-1}$ $I_n = \frac{(1-x)^n}{n} + I_{n-1}$	3 Marks Correct solution 2 Marks Correct substitution and establishes reduction formula 1 Mark Correct integration by parts
Q15 c)	$ax^3 + bx^2 + cx + d = 0, \text{ let the roots be, } \alpha, \frac{1}{\alpha}, \beta$ $\alpha + \frac{1}{\alpha} + \beta = -\frac{b}{a} \quad (1)$ $\alpha \times \frac{1}{\alpha} + \alpha \times \beta + \frac{1}{\alpha} \times \beta = \frac{c}{a}$ $1 + \beta \left(\alpha + \frac{1}{\alpha}\right) = \frac{c}{a} \quad (2)$ $\alpha \times \frac{1}{\alpha} \times \beta = -\frac{d}{a}$ $\beta = -\frac{d}{a} \quad (3)$ Sub (3) into (1) $\alpha + \frac{1}{\alpha} - \frac{d}{a} = -\frac{b}{a}$ $\alpha + \frac{1}{\alpha} = \frac{d-b}{a} \quad (4)$ Sub (3) and (4) into (2) $1 - \frac{d}{a} \left(\frac{d-b}{a}\right) = \frac{c}{a}$ $a^2 - d^2 + bd = ac$ $a^2 - d^2 = ac - bd$	3 Marks Correct solution 2 Marks Obtains $\alpha + \frac{1}{\alpha} = \frac{d-b}{a}$ 1 Mark Obtains $\beta = -\frac{d}{a}$
Q15 d) i)	In $\triangle EFP$ and $\triangle PFT$, $\angle EFP = \angle PFT$ (common) $\angle PTF = \angle TAE$ (alternate segment with PT tangent) $\angle TAE = \angle EPF$ (alternate angles, $TA \parallel PS$) $\therefore \angle PTF = \angle EPF$ $\therefore \triangle EFP$ is similar to $\triangle PFT$ (equiangular)	2 Marks Correct solution 1 Mark Proves $\angle PTF = \angle TAE$ with correct geometric reasoning

Q15 d) ii)	Since $\triangle EFP$ is similar to $\triangle PFT$, then the corresponding sides are in the same ratio. $\frac{PF}{FT} = \frac{EF}{PF}$ $\therefore PF^2 = EF \times FT$	1 Mark Correct solution
Q15 d) iii)	$FS^2 = EF \times FT$ (product of two intervals from the points to the circle on the secant is equal to the square of the tangent) $PF^2 = EF \times FT$ (shown above) $FS^2 = PF^2$ $FS = PF$ $\therefore F$ is the midpoint of PS	2 Marks Correct solution 1 Mark Proves $FS^2 = EF \times FT$ with the correct geometric reasoning
Q16 a)	The letters 'E' occurs twice in TELEGRAPH Case 1: No 'E', consider the letters TLGRAPH Number of selections is 7C_4 and the possible arrangements of this selection is $4!$ Number of arrangement with no 'E' is ${}^7C_4 \times 4! = 840$ Case 2: 1 'E' so 3 other letters from the remaining 7 letters is 7C_3 and the arrangements of this selection is $4!$ Number of arrangement with 1 'E' is ${}^7C_3 \times 4! = 840$ Case 3: 2 'E's so 2 other letters from the remaining 7 letters is 7C_2 and the arrangements of this selection is $\frac{4!}{2!}$ Number of arrangement with 2 'E's is ${}^7C_2 \times \frac{4!}{2!} = 252$ Total number of distinct arrangements = $840 + 840 + 252 = 1932$	2 Marks Correct solution 1 Mark Makes some progress
Q16 b) i)	$(m - n)^2 \geq 0$ $m^2 - 2mn + n^2 \geq 0$ $m^2 + 2mn + n^2 \geq 2mn + 2mn$ $(m + n)^2 \geq 4mn$ $m + n \geq 2\sqrt{mn} \quad (\because m > 0, n > 0)$	1 Mark Correct solution
Q16 b) ii)	$m + n \geq 2\sqrt{mn}$ $n + p \geq 2\sqrt{np}$ $p + q \geq 2\sqrt{pq}$ $(m + n)(n + p)(m + p) \geq 2\sqrt{mn} \times 2\sqrt{np} \times 2\sqrt{pq}$ $(m + n)(n + p)(m + p) \geq 8\sqrt{m^2 n^2 p^2}$ $(m + n)(n + p)(m + p) \geq 8mnp$	2 Marks Correct solution 1 Mark Makes significant progress
Q16 b) iii)	$\left(\frac{m}{n} + \frac{n}{p}\right) + \left(\frac{p}{q} + \frac{q}{m}\right) \geq 2\sqrt{\frac{m}{n} \times \frac{n}{p}} + 2\sqrt{\frac{p}{q} \times \frac{q}{m}}$ $\left(\frac{m}{n} + \frac{n}{p}\right) + \left(\frac{p}{q} + \frac{q}{m}\right) \geq 2\sqrt{\frac{m}{p}} + 2\sqrt{\frac{p}{m}}$	2 Marks Correct solution 1 Mark Makes significant progress

	$\left(\frac{m}{n} + \frac{n}{p}\right) + \left(\frac{p}{q} + \frac{q}{m}\right) \geq 2 \left(\sqrt{\frac{m}{p}} + \sqrt{\frac{p}{m}} \right)$ $\left(\frac{m}{n} + \frac{n}{p}\right) + \left(\frac{p}{q} + \frac{q}{m}\right) \geq 2 \left(2 \sqrt{\sqrt{\frac{m}{p}} \times \sqrt{\frac{p}{m}}} \right)$ $\left(\frac{m}{n} + \frac{n}{p}\right) + \left(\frac{p}{q} + \frac{q}{m}\right) \geq 2 \left(2 \sqrt{\sqrt{\frac{mp}{mp}}} \right)$ $\left(\frac{m}{n} + \frac{n}{p}\right) + \left(\frac{p}{q} + \frac{q}{m}\right) \geq 4(\sqrt{1})$ $\left(\frac{m}{n} + \frac{n}{p}\right) + \left(\frac{p}{q} + \frac{q}{m}\right) \geq 4$	
Q16 c)	$f(x) = x^{x^2}$ $f(x) = (e^{\log_e x})^{x^2}$ $f(x) = e^{x^2 \log_e x}$ Let $g(x) = x^2 \log_e x$ $f(x) = e^{g(x)}$ $f'(x) = g'(x)e^{g(x)}$ $g'(x) = 2x \log_e x + x^2 \times \frac{1}{x}$ $g'(x) = 2x \log_e x + x$ $f'(x) = (2x \log_e x + x)e^{x^2 \log_e x}$ $f'(x) = x(2 \log_e x + 1)x^{x^2}$ $f'(x) = (2 \log_e x + 1)x^{x^2+1}$	2 Marks Correct solution 1 Mark Makes significant progress
Q16 d) i)	$\frac{A^{n+1} - A^n B + B^{n+1} - B^n A}{A - B}$ $= \frac{A^n(A - B) + B^n(B - A)}{A - B}$ $= \frac{A^n(A - B) - B^n(A - B)}{A - B}$ $= \frac{(A - B)(A^n - B^n)}{A - B}$ $= A^n - B^n$	1 Mark Correct solution
Q16 c) ii)	$A^{n+1} + B^{n+1} = (A - B)(A^n - B^n) + A^n B + B^n A$ Given that $A > 0, B > 0$, If $A \geq B$, then $A - B \geq 0$ and $A^n - B^n \geq 0$, So $(A - B)(A^n - B^n) \geq 0$ If $A < B$, then $A - B < 0$ and $A^n - B^n < 0$, So $(A - B)(A^n - B^n) > 0$ $\therefore (A - B)(A^n - B^n) \geq 0$ Hence $A^{n+1} + B^{n+1} \geq 0 + A^n B + B^n A$ $\therefore A^{n+1} + B^{n+1} \geq A^n B + B^n A$	2 Marks Correct solution 1 Mark Recognises and establishes relationship from part i)
Q16 c) iii)	1. Prove statement is true for $n = 1$. $LHS = \left(\frac{A+B}{2}\right)^1$ $LHS = \frac{A+B}{2}$	3 Marks Correct solution

	$RHS = \frac{A^1 + B^1}{2}$ $RHS = \frac{A + B}{2}$ $LHS = RHS$ $\therefore$ statement is true for $n = 1$ 2. Assume statement is true for $n = k$ (k some positive integer) i.e. $\left(\frac{A + B}{2}\right)^k \leq \frac{A^k + B^k}{2}$ 3. Prove statement is true for $n = k + 1$ i.e. $\left(\frac{A + B}{2}\right)^{k+1} \leq \frac{A^{k+1} + B^{k+1}}{2}$ $\left(\frac{A + B}{2}\right)^{k+1} = \left(\frac{A + B}{2}\right)^k \times \left(\frac{A + B}{2}\right)$ From step 2 $\left(\frac{A + B}{2}\right)^{k+1} \leq \frac{A^k + B^k}{2} \times \left(\frac{A + B}{2}\right)$ $\left(\frac{A + B}{2}\right)^{k+1} \leq \frac{A^{k+1} + A^k B + B^k A + B^{k+1}}{4}$ $\left(\frac{A + B}{2}\right)^{k+1} \leq \frac{A^{k+1} + B^{k+1}}{4} + \frac{A^k B + B^k A}{4}$ From part ii) $\left(\frac{A + B}{2}\right)^{k+1} \leq \frac{A^{k+1} + B^{k+1}}{4} + \frac{A^{k+1} + B^{k+1}}{4}$ $\left(\frac{A + B}{2}\right)^{k+1} \leq \frac{2(A^{k+1} + B^{k+1})}{4}$ $\left(\frac{A + B}{2}\right)^{k+1} \leq \frac{A^{k+1} + B^{k+1}}{2}$ $\therefore$ Statement is true by mathematical induction for all positive integer n.	2 Marks Makes significant progress 1 Mark Prove statement is true for $n = 1$.
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